Hyungseok Kim

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Research Interests

Quantum Compilers, Quantum Error Correction, Quantum Programming, Quantum Algorithms.

Education

Yonsei University Seoul, Korea
M.S. in Electrical and Electronic Engineering Expected Feb. 2027
- Advisor: Prof. Won Woo Ro
- Thesis: "Efficient Lattice Surgery Compilation for Fault Tolerant Quantum Computing" — Based on [UR2].

Yonsei University Seoul, Korea
B.S. in Electrical and Electronic Engineering Feb. 2025
- Capstone Project: *HS-Table: A Table-Based Algorithm for Optimizing Quantum Circuits through Qubit Reuse* (Advisor: Prof. Won Woo Ro) — Expanded and published as [C3] in ISCA '25.

B.A. in Economics (Double Major) Feb. 2025

Research Positions

Yonsei University Seoul, Korea
- Graduate Researcher, *Embedded Systems and Computer Architecture Lab* (Advisor: Prof. Won Woo Ro) Mar. 2025 – Present
- Undergraduate Researcher, *Embedded Systems and Computer Architecture Lab* (Advisor: Prof. Won Woo Ro) Jun. 2024 – Feb. 2025

Publications (Google Scholar)

In Preparation

[IP1] Youngmin Kim, Enhyeok Jang, **Hyungseok Kim**, Changheon Lee, and Won Woo Ro. "Parameter-Freezing Variational Quantum Imaginary Time Evolution with Reduced Measurement and Computational Overhead."

Under Review

[UR3] Changheon Lee, **Hyungseok Kim**, Sungho Pyun, Enhyeok Jang, Hanbit Lee, and Won Woo Ro. "Bell Pair Buffering for Remote Stabilizer Measurement."

[UR2] **Hyungseok Kim**, Changheon Lee, Seungjik Kim, Enhyeok Jang, Youngmin Kim, Seungwoo Choi, Hanbit Lee, Sungho Pyun, and Won Woo Ro. "SpiderLS: Leveraging Full ZX Reduction for Lattice Surgery Compilation." *arXiv preprint*, 2026. arXiv:2608.30228
  * Artifact available at https://github.com/fluorite42/SpiderLS

[UR1] Enhyeok Jang, Changheon Lee, Seungjik Kim, Jain Koo, Sungho Pyun, **Hyungseok Kim**, and Won Woo Ro. "tripleS-KQD: A Shot-Scaled Sample-Based Krylov Diagonalization."

Conference Papers

[C6] Sungho Pyun, Changheon Lee, **Hyungseok Kim**, Enhyeok Jang, Youngmin Kim, and Won Woo Ro. "DARTH: Lookahead-Driven Compilation Exploiting Transient Qubits for Distributed Quantum Computing." *The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO '26)*, Athens, Greece, 2026. DOI:10.1109/MICRO67795.2026.00208

[C5] Enhyeok Jang, **Hyungseok Kim**, Yongju Lee, Jaewon Kwon, Yipeng Huang, and Won Woo Ro. "Toward Scalable Gate-Level Parallelism on Trapped-Ion Processors with Racetrack Electrodes." *2026 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, Sydney, NSW, Australia, 2026. DOI:10.1109/HPCA68181.2026.11408608 arXiv:2601.08930

* Acknowledgment: Dr. Zichang He (JPMorganChase) and Tianyi Hao (University of Wisconsin-Madison) for discussions on QAOA runtime estimation.

[C4] Changheon Lee, **Hyungseok Kim**, Seungwoo Choi, Youngmin Kim, and Won Woo Ro. "d'ArQ: A QOC Framework with Causality-Aware Grouping and Basis Selection." *2026 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, Sydney, NSW, Australia, 2026. DOI:10.1109/HPCA68181.2026.11408534

[C3] **Hyungseok Kim**, Enhyeok Jang, Seungwoo Choi, Youngmin Kim, and Won Woo Ro. "QR-Map: A Map-Based Approach to Quantum Circuit Abstraction for Qubit Reuse Optimization." *The 52nd Annual International Symposium on Computer Architecture (ISCA '25)*, Tokyo, Japan, 2025. DOI:10.1145/3695053.3731020

- Merit Academic Paper Award, *Yonsei University Graduate School*, 2026.

[C2] Enhyeok Jang, Youngmin Kim, **Hyungseok Kim**, Seungwoo Choi, Yipeng Huang, and Won Woo Ro. "Qubit Movement-Optimized Program Generation on Zoned Neutral Atom Processors." *The 23rd ACM/IEEE International Symposium on Code Generation and Optimization (CGO '25)*, Las Vegas, NV, USA, 2025. DOI:10.1145/3696443.3708937 arXiv:2503.02272

* Acknowledgment: Dr. Hengyun Zhou (QuEra Computing) for discussions on real zoned neutral atom processor, Prof. Joonhee Choi (Stanford University) for discussions on tweezer array, and Young Jung Kang (Yonsei University) for discussions on quantum error correction.

[C1] Enhyeok Jang, Dongho Ha, Seungwoo Choi, Youngmin Kim, Jaewon Kwon, Yongju Lee, Sungwoo Ahn, **Hyungseok Kim**, and Won Woo Ro. "Recompiling QAOA Circuits on Various Rotational Directions." *International Conference on Parallel Architectures and Compilation Techniques (PACT '24)*, Long Beach, CA, USA, 2024. DOI:10.1145/3656019.3676899

Journal Papers

[J1] Youngmin Kim, Enhyeok Jang, **Hyungseok Kim**, Seungwoo Choi, Changheon Lee, Donghwi Kim, Woomin Kyoung, Kyujin Shin, and Won Woo Ro. "Distribution-Adaptive Dynamic Shot Optimization for Variational Quantum Algorithms." *Physical Review Research*, 7(4), 043253, 2025. DOI:10.1103/vwhk-22b8 arXiv:2412.17485

* Acknowledgment: Dr. Seung Hyo Noh (Hyundai Motor Company) for enlightening discussions.

Workshop & Non-Archival Presentations

[NC3] **Hyungseok Kim** and Won Woo Ro. "Graph-Based Compilation for Lattice Surgery." *The 26th International Conference on Electronics, Information, and Communication (ICEIC '27)*, Fukuoka, Japan, 2027. (Abstract)

* Invited by Prof. Sunghye Park (Ajou University) to Special Session: Emerging Computing Technologies.

[NC2] Junwoo You, **Hyungseok Kim**, Minchan Kim, and Won Woo Ro. "Profiling GPU-SSD I/O Behavior for Large Language Model Inference." *International Conference on AI for Science 2026 (AI4Sci Korea '26)*, Seoul, Korea, 2026. (Abstract)

[NC1] Changheon Lee, Enhyeok Jang, Youngmin Kim, **Hyungseok Kim**, Sungho Pyun, and Won Woo Ro. "TENET: A Pincer Movement with Backward Programs for Bypassing Relaxed Qubit Readouts." *The 25th International Conference on Electronics, Information, and Communication (ICEIC '26)*, Macau SAR, China, 2026.

* Acknowledgment: Dr. Ritajit Majumdar (IBM Quantum) for discussions on the behavior of IBM machines and Sir Christopher Edward Nolan (Directors Guild of America (DGA)) for inspiring the research idea.

[W2] **Hyungseok Kim**, Enhyeok Jang, Youngmin Kim, and Won Woo Ro. "Native Gate-Aware QAOA Ansatz." *The 1st International Workshop on Quantum Data and Machine Learning: Systems, Theory and Hardware (QDML '25)*, In conjunction with ICDE '25, Hong Kong SAR, China, 2025. (Abstract)

[W1] Enhyeok Jang, Youngmin Kim, **Hyungseok Kim**, Dongho Ha, Yongju Lee, Jaewon Kwon, Jun Woo You, Jiho Park, and Won Woo Ro. "A Dead Gate Elimination for Quantum Programs." *The 1st HPC/AI Integration with Quantum Computing Workshop (HAIQ '25)*, In conjunction with HPCA '25, Las Vegas, NV, USA, 2025.

Domestic Papers

[DC2] Sungho Pyun, Changheon Lee, **Hyungseok Kim**, Enhyeok Jang, Youngmin Kim, and Won Woo Ro. "Dynamic-Mapping Compilation Using Transient Qubits for Distributed Quantum Computing." *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Seogwipo, Korea, 2026. (In Korean)

- Industry Interest Paper Award (LG Electronics), *The Institute of Electronics and Information Engineers (IEIE)*, 2026.

[DC1] Changheon Lee, Youngmin Kim, **Hyungseok Kim**, and Won Woo Ro. "How Basis Choice Shapes Performance in Quantum Optimal Control Framework." *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Seogwipo, Korea, 2025.

[DM1] Youngmin Kim, Changheon Lee, **Hyungseok Kim**, and Won Woo Ro. "Research Trends and Prospects of Hybrid Computing-Based Variational Quantum Algorithms (VQAs)." *The Magazine of the Institute of Electronics and Information Engineers (IEIE)*, 52(9), 2025. (Magazine article, in Korean)

R&D Projects

University-Industry Foundation (UIF), Yonsei University Seoul, Korea

- Research Assistant, Optimizing GPU-SSD Integrated System Architecture for Large-Scale Artificial Intelligence Learning, *Samsung Electronics DS Division* Oct. 2025 – Sep. 2026
- Research Assistant, Research on Quantum Pulse Latency Reduction Through Quantum Optimal Control Framework, *National Research Foundation of Korea (NRF)* Jan. 2025 – Dec. 2025
- Research Assistant, Developing a High-Speed Quantum Circuit Simulator Supports Verification and Testing of Hyperscale Quantum Algorithms, *Hyundai NGV* Jun. 2024 – Oct. 2024

Professional Service

Organizer

- Tutorial Co-Organizer, SpaceQat: Emulating Multi-Levels of Quantum Dynamics via Single Program, In conjunction with HPC Asia '27 Feb. 2027

Reviewer

- Future Generation Computer Systems (FGCS) (SCIE Q1, JCR 2025) Oct. 2026
- The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO '26) Artfact Evaluation Aug. 2026

Sub-Reviewer

- The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO '26) May 2026
- The 53rd International Symposium on Computer Architecture (ISCA '26) Jan. 2026
- The 52nd International Symposium on Computer Architecture (ISCA '25) Jan. 2025

Student Volunteer

- The 58th IEEE/ACM International Symposium on Microarchitecture (MICRO '25) Oct. 2025

Laboratory Website Administrator

- Embedded Systems and Computer Architecture Lab, *Yonsei University* Mar. 2025 – Feb. 2027

Scholarships and Awards

Fellowships and Scholarships

- BK (Brain Korea) 21 Research Fellowship, *Yonsei University* 2025 – 2026
- Graduate Student Research Assistant (GSRA) Scholarship (New Student Excellence), *Yonsei University* 2025

Honors and Awards

- Industry Interest Paper Award (LG Electronics), *The Institute of Electronics and Information Engineers (IEIE)* 2026
- Merit Academic Paper Award, *Yonsei University Graduate School* 2026
- Student-Designed Major Competition, *Yonsei University* 2nd Place (with $360) 2024

- Academic Honors (Top 10%), *Yonsei University* 2024
- Micro-Learning Video Contest, *Yonsei University* 4th Place (with $380) 2023
- Academic Honors (Top 10%), *Yonsei University* 2022

Teaching Experience

Teaching Assistant, Yonsei University
- EEE 4473: Embedded System Lab. (Undergraduate) Spring 2025
- EEE 4610: Electrical and Electronic Engineering Capstone Design (Undergraduate) Spring 2025

Mentoring
- Research Mentor for Two Undergraduate Students, *Quantum Information Science Club Association (QISCA)* Nov. 2025 – Jan. 2026
- Academic Mentor for Two High School Students, *Dong Incheon High School* Mar. 2019 – Jun. 2019

Talks

Invited Talks

"SpiderLS: Leveraging Full ZX Reduction for Lattice Surgery Compilation"
- *Topological Quantum Error Correction (TQEC) Design Automation* Sep. 2026
  * Invited by Dr. Austin G. Fowler (Founder of TQEC, Formerly at Google Quantum AI)

"QR-Map: A Map-Based Approach to Quantum Circuit Abstraction for Qubit Reuse Optimization"
- Journal Club, *Quantum Information Science Club Association (QISCA)* Sep. 2025
  * Invited by SungBin Lee (Founder of QISCA, Seoul National University)

Contributed Talks

"Demo 2: Transmon Emulation"
- SpaceQat: Emulating Multi-Levels of Quantum Dynamics via Single Program, *HPC Asia '27* Feb. 2027

"Advanced LLM Systems: Reasoning Model"
- Large Language Models (LLMs) Seminar, *Yonsei University* Feb. 2026

"Advanced Techniques for Instruction Delivery and Speculation & Cross-Cutting Issues"
- Advanced Computer Architecture Seminar, *Yonsei University* Feb. 2025

"Understanding Interrupt Signals in Linux Kernel"
- Advanced Operating Systems Seminar, *Yonsei University* Jan. 2025

Selected Coursework

Yonsei University, Graduate, *Electrical and Electronic Engineering*
- *Quantum Computing*: Quantum Computer and Simulation 1, Quantum Computing, Quantum Data Science, Advanced Quantum Machine Learning
- *Computer Architecture*: Compiler Design and Optimization, Advanced Programming
- *Machine Learning*: Special Topics in Pattern Recognition
- *Digital System Design*: Semiconductor Memory Design

Yonsei University, Undergraduate, *Electrical and Electronic Engineering*
- *Quantum Computing*: Quantum Machine Learning
- *Computer Architecture*: Computer Architecture, Communication Networks, Operating Systems, Microprocessors, Multicore and GPU Programming
- *Machine Learning*: Introduction to Artificial Intelligence, Intelligent Control
- *Digital System Design*: Digital Microelectronics, System IC Design, Embedded System Lab.

Yonsei University, Undergraduate, *Economics*
- *Econometrics*: Econometrics I, Financial Econometrics
- *DSGE Modeling*: Dynamic Analysis of Macroeconomy, Quantitative Macroeconomics
- *Financial Engineering*: Introduction to Financial Engineering, AI Assisted Financial Engineering

Skill Set

Technical Qualifications

- Adobe Certified Professional (ACP) in Digital Video Using Adobe Premiere Pro, *Adobe Inc.* 2024
- General Computer User, Level-I (Highest), *Korea Chamber of Commerce and Industry* 2020

Programming Languages: C/C++ (OpenMP, CUDA), Python (Qiskit), JavaScript, MATLAB, Assembly Languages (RISC-V, ARM)

Other Languages and Tools: Verilog, LaTeX, Markdown, HTML, YAML, CSS, Jekyll, Git

Languages

Working Proficiency: Korean (Native), English (CEFR C1, TOEFL iBT 5.0/6.0)

In Progress: Spanish, Chinese, Japanese

Extracurricular Activities

Professional Affiliations

- Graduate Student Member, *Institute of Electrical and Electronics Engineers (IEEE)* 2025 – Present
- Trainee, *Automotive System IC Fusion Human Resource Research Center (ASRC)* 2025 – 2026
- Trainee, *Korea Collaborative & High-tech Initiative for Prospective Semiconductor research (K-CHIPS)* 2026

Student Activities

- Participant, IonQ Research Mentoring Program, *IonQ* 2025
- Participant, Language Exchange Program, *Yonsei Korean Language Institute (KLI)* 2023 – 2024
  - 5 Quarters: 4 English–Korean, 1 Spanish–Korean
- Participant, Virtual Pen Pal Program, *University of South Florida* 2023
- Participant, Waseda-Yonsei Campus & Cultural Exchange, *Waseda University* 2019

Military Service

- Personnel Administrator (Sgt.), *142nd Intelligence Battalion, II Corps, Republic of Korea Army* 2020 – 2022

Miscellaneous

- Technical Blog: fluorite42.github.io/categories
  - "An Introduction to Quantum Error Correction" (5-part series) 2025
  - "A Guide to Quantum Gates" (2-part series) 2025
- Independent Global Traveler: Backpacked 48 Cities across 30 Countries 2023 – Present
- Erdős Number: 4